Prairie Pulse E-Bikes - Reconciliation and Validation Report

VisiVault Analytics Ltd. · October 2026 · Financial module (Phases 12-18) of the Prairie Pulse E-Bikes proof of concept

SIMULATED FINANCIAL INFORMATION - illustrative Calgary management accounts, not Prairie Pulse E-Bikes' actual financials. This report validates the internal consistency of simulated accounts; it is not an audit or review of Prairie Pulse E-Bikes' financial information.

Result: 10 of 10 mandatory checks PASS. 103,614 general-ledger lines in 19,894 journal entries; total debits $50,022,181.75 = total credits $50,022,181.75.

1. Mandatory validation checks

#	Mandatory check	Result	Evidence
1	Total debits equal total credits (every journal entry and the whole ledger)	PASS	19,894 journal entries / 103,614 lines; total debits $50,022,181.75 = credits $50,022,181.75; unbalanced entries: 0
2	Trial balances reconcile (monthly TB balances; TB P&L = income statement; AR/AP sub-ledgers = control accounts)	PASS	34 monthly trial balances (Dec-2023 to Sep-2026) all balance; TB net income ties to IS for FY2024, FY2025, YTD-2026; 6 control accounts tie to open-item sub-ledgers every month
3	Assets = Liabilities + Equity (every historical month-end and every forecast month, all scenarios)	PASS	Historical max |A-L-E| = $0.00 over 33 month-ends + opening; forecast max = $0.000000 across 3 scenarios
4	Net income flows into retained earnings (year-end closing entries; current-period earnings = YTD net income)	PASS	RE Dec-24 = opening RE -63,638.88 + FY24 NI -10,840.94; RE Dec-25 = RE Dec-24 + FY25 NI -23,731.75; current-period earnings Sep-26 = YTD NI 87,122.97; forecast Dec-26 close also ties
5	Cash-flow closing cash = balance-sheet cash (monthly, quarterly, annual, forecast); indirect = direct method	PASS	Indirect-method closing cash ties to GL cash in all 33 months, 11 quarters, 3 years and 36 forecast months; direct-method net change equals indirect every month
6	Inventory balances reconcile with inventory movement records	PASS	GL 1200 = sum of movements x standard cost at every month-end (max diff $0.00); closing units 378 = product master 378
7	Revenue and cost of goods sold reconcile to operational transactions	PASS	14 revenue/COGS lines x 33 months tie to sales, returns, service and movement records; operational-P&L-to-GL revenue and EBIT bridges have zero residual
8	Financial forecasts remain integrated when assumptions change	PASS	60 random assumption sets (demand 0.6-1.4x, margins +/-5 pts, costs -10/+15%, DPO +/-20 days, rates -2/+4 pts, cover 0.6-1.5x): max |A-L-E| or |cash diff| = $0.000000; Excel recalculation of the formula model (Base; Conservative; Custom): max |Excel - Python| = $0.0000, Excel checks TRUE = True

#	Mandatory check	Result	Evidence
9	No financial statement contains unexplained balancing plugs	PASS	No suspense/plug/rounding accounts exist; every line posts to the chart of accounts from a documented journal source; every balance-sheet account maps to one cash-flow line; opening balances come from explicit start-up entries; operational-to-GL bridges have zero residual; the forecast has no circular references (interest on opening balances) and balances by construction of complete balance-sheet roll-forwards
10	All synthetic financial information is prominently labelled	PASS	35 CSV tables carry data_status; 43 sheets in 5 workbooks carry the SIMULATED banner; 3 reports labelled

2. Journal entries by source

Every entry is generated from an operational record or a documented policy; none is a balancing entry.

source	entries	lines	debits	credits
Capital expenditure	5	15	58,800.00	58,800.00
Collections	1,556	3,112	8,216,568.74	8,216,568.74
Depreciation	33	66	101,573.25	101,573.25
Financing	91	182	1,311,471.30	1,311,471.30
GST remittance	10	30	412,925.39	412,925.39
Head office	10	20	19,500.00	19,500.00
Income tax	4	8	4,212.16	4,212.16
Inventory adjustment	74	148	30,282.98	30,282.98
Opening balance	24	55	1,376,336.81	1,376,336.81
Operating expenses	264	726	1,603,064.37	1,603,064.37
Payroll	98	229	2,669,072.37	2,669,072.37
Prepaid insurance	36	72	164,849.28	164,849.28
Purchasing	1,719	5,150	5,961,799.02	5,961,799.02
Returns	474	1,185	371,043.75	371,043.75
Sales	5,438	41,021	12,668,973.85	12,668,973.85
Service	7,735	46,895	3,132,237.50	3,132,237.50
Supplier payments	2,321	4,642	5,770,570.44	5,770,570.44
Year-end close	2	58	6,148,900.54	6,148,900.54

3. Trial balances

Pre-closing trial balance totals at selected month-ends (all 34 monthly trial balances balance).

month	debits	credits	difference
2023-12	910,514.96	910,514.96	0.00
2024-03	1,620,855.54	1,620,855.54	0.00
2024-06	2,702,173.10	2,702,173.10	0.00
2024-12	4,068,061.25	4,068,061.25	0.00
2025-03	1,330,333.40	1,330,333.40	-0.00
2025-06	2,762,629.33	2,762,629.33	0.00
2025-12	4,019,525.85	4,019,525.85	0.00
2026-03	1,762,573.09	1,762,573.09	0.00
2026-06	2,928,458.71	2,928,458.71	0.00
2026-09	3,947,588.39	3,947,588.39	0.00

4. Subsidiary ledgers vs control accounts

Open-item sub-ledgers (receivables by tender counterparty / customer / manufacturer; payables by supplier and vendor) at every month-end.

ledger	control_account	account_name	months	balance_30sep2026	max_abs_difference
AP	2000	Accounts payable - inventory & parts suppliers	33	157,952.65	0.00
AP	2010	Accounts payable - operating vendors	33	27,972.25	0.00
AR	1100	Card & debit processor settlements receivable	33	13,241.55	0.00
AR	1110	Consumer-financing partner receivable	33	6,713.68	0.00
AR	1120	Trade receivables - business/fleet accounts	33	10.49	0.00
AR	1130	Manufacturer warranty claims receivable	33	2,758.50	0.00

5. Inventory reconciliation

GL account 1200 vs the independent valuation of inventory movement records (opening + receipts - sales + returns - adjustments, at standard cost).

month	units_on_hand	value_from_movements	gl_1200	difference
2024-03	585	678,710.31	678,710.31	0.00
2024-06	339	317,051.88	317,051.88	0.00
2024-09	451	389,634.13	389,634.13	0.00
2024-12	437	410,467.71	410,467.71	0.00
2025-03	442	420,183.86	420,183.86	0.00
2025-06	294	238,496.44	238,496.44	0.00
2025-09	378	307,577.71	307,577.71	0.00
2025-12	434	415,275.11	415,275.11	0.00
2026-03	577	652,730.09	652,730.09	0.00
2026-06	305	317,073.75	317,073.75	0.00
2026-09	378	342,554.88	342,554.88	0.00

6. Revenue and cost of goods sold vs operational transactions

Jan 2024 - Sep 2026 totals; maximum monthly difference across 33 months.

account	line	operational	general_ledger	max_monthly_difference
4000	4000 Bicycle sales (gross)	6,560,030.99	6,560,030.99	0.00
4100	4100 Accessories sales (gross)	494,833.33	494,833.33	0.00
4110	4110 Batteries & chargers (gross)	185,601.19	185,601.19	0.00
4120	4120 Parts (gross)	43,942.76	43,942.76	0.00
4200	4200 Service labour (customer)	942,379.00	942,379.00	0.00
4210	4210 Service parts revenue	706,368.52	706,368.52	0.00
4220	4220 Warranty reimbursements	42,665.40	42,665.40	0.00
4900	4900 Discounts & allowances	169,744.16	169,744.16	0.00
4910	4910 Returns & refunds	210,960.10	210,960.10	0.00
5000	5000 COGS e-bikes (sales cost less returns restocked)	4,481,605.73	4,481,605.73	0.00
5010	5010 COGS accessories/batteries/parts	397,690.66	397,690.66	0.00
5100	5100 Shrinkage & write-offs (movement records)	30,282.98	30,282.98	0.00
5200	5200 Service parts cost (completed orders)	382,450.95	382,450.95	0.00
5210	5210 Direct service labour (completed orders)	281,622.32	281,622.32	0.00

7. Bridges from the pre-existing operational P&L to the general ledger

Net revenue (Jan 2024 - Sep 2026)

item	cad
Total revenue per operational P&L (completed retail lines + all service orders by intake date)	8,614,336.98
+ Net sales of lines later returned (operational P&L excludes them)	210,960.10
- Refunds issued (GL contra-revenue 4910)	-210,960.10
- Revenue of service orders still in progress at 30-Sep-2026 (GL recognises on completion)	-19,220.05
= Explained GL net revenue	8,595,116.93
GL net revenue Jan-2024 to Sep-2026	8,595,116.93
Unexplained residual (must be 0)	0.00

Operating profit to EBIT by fiscal year

Each reconciling item is computed from source data, not from the ledger; the residual is the unexplained difference.

item	2024-01 to 2024-12	2025-01 to 2025-12	2026-01 to 2026-09
Operating profit per operational P&L (monthly_pnl, pre-existing dashboards)	27,399.91	17,641.45	127,906.79
+ Imputed inventory financing cost removed (replaced by actual interest below EBIT)	31,577.47	19,567.92	16,840.41
+/- Retail returns timing: GP of later-returned lines sold in window less GP reversed by refunds in window	1,046.28	-495.08	-551.20
- Inventory shrinkage & damaged write-offs (movement records; not in operational P&L)	-12,148.15	-11,696.39	-6,438.44
+/- Service recognition timing: GP of orders completed in window less GP of orders received in window	-2,492.73	771.78	-8,835.30
- Depreciation of fixed assets (not in operational P&L)	-33,366.64	-36,277.26	-31,929.35
= Explained GL EBIT	12,016.14	-10,487.58	96,992.91
GL EBIT (income statement)	12,016.14	-10,487.58	96,992.91
Unexplained residual (must be 0)	0.00	-0.00	0.00

8. Forecast integration

scenario	max_abs_bs_check	max_abs_cash_check
Conservative	0.000000	0.000000
Base	0.000000	0.000000
Optimistic	0.000000	0.000000

Random-assumption stress test: 60 assumption sets; max |A - L - E| and |CF closing cash - BS cash| below $0.01 in every month.

Excel recalculation of the formula model

Excel was not available on the build machine; the workbook formulas are generated from the same templates the Python engine evaluates.

9. Cash control

Lowest end-of-day bank balance $20,099 on 2024-04-02; days with a negative balance: 0.

10. Labelling

35 CSV tables carry data_status; 43 sheets in 5 workbooks carry the SIMULATED banner; 3 reports labelled

Reproduce: `.venv/bin/python run_pipeline.py --from finance` then `.venv/bin/python -m pytest tests/test_accounting.py`.